

Mollic Gleysols

Mollic Gleysols are hydromorphic mineral soils formed under conditions of high groundwater influence or poor drainage. They are characterized by a thick, dark, organic-rich mollic horizon (surface 20–40 cm), which gives them high fertility and good structure.

The term “Gleysol” indicates the presence of reduction features such as mottling, grey or blue-grey colors in the subsoil due to periodic water saturation and anaerobic conditions. These soils typically develop on fine-textured alluvial or lacustrine deposits in low-lying landscapes.

Drainage:

- **Wet season:** Poorly drained; waterlogging is common due to shallow water tables and fine texture.
- **Dry season:** Drainage improves slowly, but the soil may remain moist for long periods.

Landscape: Common on river valleys, floodplains, depressions, and shallow basins with high water tables. Found in parts of western and coastal Kenya, along rivers, lakeshores, and deltaic deposits.

Depth: Moderately deep to deep (60–150 cm); rooting may be restricted in areas with prolonged saturation or gleyed subsoil layers.

Workability:

- **When wet:** Sticky and plastic; difficult to cultivate due to waterlogging.
- **When dry:** Firm but workable; cracking may occur in clay-rich subsoils. Organic matter incorporation improves tilth.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Moderate to high (15–25 cmol(+)/kg), depending on clay content and organic matter.
- **Base saturation:** High (>60%), giving good natural fertility.
- **Nutrients:** Nitrogen and organic matter are abundant in the mollic horizon; phosphorus may be limiting due to fixation in clay minerals; potassium, calcium, and magnesium generally adequate.
- **Organic matter:** High (3–6%) in the surface mollic horizon, contributing to fertility, structure, and water retention.

pH:

Slightly acidic to neutral (5.5–7.0), depending on drainage, organic matter, and parent material.

Management:

- Avoid heavy machinery during wet periods to prevent compaction.
- Use raised beds or drainage channels to improve aeration and root development.
- Maintain organic matter levels through manure, compost, and green manure.
- Apply phosphorus fertilizers carefully due to fixation in clay soils.
- Implement crop rotation and cover cropping to enhance fertility and soil structure.

Suitable Crops:

Highly productive for cereals (maize, rice, wheat), legumes (beans, cowpeas), vegetables (tomatoes, onions, cabbages), and forage crops. With proper drainage, irrigation, and fertility management, these soils are very suitable for intensive agriculture.